RAG-Based Customer Support Assistant

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1. Introduction

1.1 Overview

The system implements a Retrieval-Augmented Generation pipeline to enhance customer support automation.

1.2 Objectives

- Improve response accuracy
- Reduce hallucinations
- Enable scalable support

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2. System Architecture

The system integrates:

- Retrieval layer (ChromaDB)
- LLM layer (Groq/Gemini)
- Workflow orchestration (LangGraph)

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3. Design Decisions

3.1 Chunk Size

- 800 characters
- Balance between context and efficiency

3.2 Embedding Strategy

- Local HuggingFace embeddings
- Reduces cost

3.3 Retrieval Approach

- Hybrid retrieval improves accuracy

3.4 Prompt Design

- Context-based answering
- Avoid hallucination

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4. Workflow Explanation

4.1 LangGraph Usage

Defines structured execution flow.

4.2 Node Responsibilities

- Retrieval: fetch relevant data
- Agent: generate response
- Evaluation: score output

4.3 State Transitions

State is updated at each node.

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5. Conditional Logic

5.1 Intent Detection

- Uses LLM classification

5.2 Routing

- Directs queries to appropriate agent

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6. HITL Implementation

6.1 Role

Handles edge cases and complex queries.

6.2 Benefits

- Improves reliability
- Handles ambiguity

6.3 Limitations

- Requires human availability

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7. Challenges & Trade-offs

Trade-off	Description
Accuracy vs Speed	Larger models slower
Chunk size vs context	Larger chunks improve context
Cost vs performance	Groq reduces cost

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8. Testing Strategy

8.1 Approach

- Functional testing
- Query validation

8.2 Sample Queries

- “What is return policy?”
- “Refund not received”

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9. Future Enhancements

- Multi-document support
- Persistent memory
- Feedback loop
- Deployment